

K701 — The FK boundary-weight exponent may be DERIVABLE (not just citable): the leptons are SPINORS, so the weight is the spinor-shifted Szegő exponent $= 5/2 - 1/2 = E_0 = 2$. Resolves Casey’s $E_0=2$ -vs- $5/2$ fork by derivation. + α closure audit (Grace DIG 2: CONDITIONAL PASS).

Keeper | 2026-07-15 | Casey named the exact fork: $E_0=2$ vs the Szegő exponent $5/2$. Elie computed both (A: conformal/boundary→ratios 3,4=the E-ladder; B: Bergman/bulk→ $11/2, 13/2$) and retracted his over-quick consistency claim (good — the fish-detector caught it). Keeper: the fork may be DERIVABLE from the field’s SPIN, which would resolve it without the FK citation. + α ’s 4π closure audited.

THE FK EXPONENT — a DERIVATION path (turns “citation we don’t have” into “forced by the spin”)

The fork is: boundary weight exponent $= E_0=2$ (conformal energy) or $5/2$ (Szegő). Here is why it should be 2, by derivation: - The Szegő exponent $5/2 = \text{genus}/2$ ($n_C/2$) is the SCALAR Hardy-space weight (the boundary weight of holomorphic *scalar* functions on D_{IV}^5 ; the overlap $N^{\{5/2\}}$). - But the leptons/quarks are SPINORS (Di spinors, K539). A spin- $1/2$ field’s boundary/Hardy weight is the scalar weight shifted by the spin: the free-Dirac-fermion conformal dimension in d dimensions is $(d-1)/2$. For $d=5$: $(5-1)/2 = 2 = E_0$. And $5/2 - 2 = 1/2 =$ exactly the spin- $1/2$ shift. - So: the fermion boundary weight $=$ the scalar Szegő exponent ($5/2$) $-$ the spinor shift ($1/2$) $= 2 = E_0$. The $5/2$ is the *scalar* weight; the matter is fermionic, so the weight is 2. This is target-innocent (it’s the field’s spin, not the observed angles) and it’s a DERIVATION, not a citation. - It’s TRIPLY consistent: (i) it equals Lyra’s E-ladder ground (electron $E_0=2 =$ the free-fermion dimension in 5D — F544); (ii) it gives Elie’s (A) climbing ratios 3, 4 = the conformal energies $E_\mu=3$, $E_\tau=4$ (target-innocent, matches the addresses); (iii) the projection prior (K697/K700, observables are boundary) already favored (A). Three independent priors, none a fit.

What this means for the finish (what’s needed / how to derive it)

- We may NOT need the FK citation. The exponent is forced by the field’s spin: spinor $\rightarrow E_0 = 2$ ($=$ the free-fermion dimension $=$ the scalar Szegő minus the spin shift). The FK citation would CONFIRM it; the spinor-shift derivation FORCES it.
- What’s needed (rank order): (1) Lyra verifies the spinor shift is exactly $-1/2$ — that the discrete-series/Hardy space for the spin- $1/2$ field on D_{IV}^5 carries exponent $(\text{genus}-1)/2 = 2$, not the scalar genus/2 $= 5/2$. This is her rep-theory lane and it’s a clean, well-posed check (the spinor bundle’s Hardy weight). (2) With $E_0=2$ fixed, Grace runs the six on the (A) ladder (positions $\{1, 2/3, 1/2\}$, ratios 3,4). (3) The FK citation, if obtained, is the confirmation — nice to have, not the gate.
- So the finish moves from “one citation we don’t have” to “one spinor-weight check Lyra can do.” That’s a derivation, not a wait.

α CLOSURE AUDIT (Grace DIG 2, K699) — CONDITIONAL PASS

Grace closed the 4π : $\alpha =$ the D_{IV}^5 shell-capacity (137, charge=integer $SO(2)$ -weight) projected to physical $3+1$ via $SO(5,2) \rightarrow SO(3,1)$; there a point charge’s flux passes through a 2-sphere of solid angle $\int d\Omega = 4\pi = \text{Vol}(S^2)$ — the 4π in $\alpha = e^2/4\pi$. - VERDICT: CONDITIONAL PASS on α ’s full closure. The 4π is now IDENTIFIED as the standard 3D Coulomb solid angle *carried by the descent* — which answers the only open question (“is the 4π a free BST knob?” $\rightarrow$ NO, it’s the descent’s 3-space solid angle). α now has no free normalization: the finite-capacity count (137) + the projection’s solid angle (4π), both geometric. - The condition (honest, referee-facing): this IDENTIFIES the 4π as Coulomb’s 3D solid angle carried by the descent; it does NOT re-derive Coulomb’s law from scratch (nor should it — Coulomb’s law in the observed 3-space is not BST’s to re-derive). So α is “fully closed” in the sense that *no free parameter remains* — the 4π is fixed geometry, not a fit. State it that way (not “we re-derived Coulomb”). - So α is complete AND its normalization is geometric ($137 =$ capacity, $4\pi =$ descent solid angle), on top of the forward curvature (n_C/N_{max} , K700). α is done.

The 26 — state

14 banked (incl. α fully closed), 6 identified-strong, 6 behind the ONE gate — now a spinor-weight CHECK ($E_0=2$) rather than a citation-we-lack. Resolve the spinor shift $\rightarrow$ run the six $\rightarrow$ the table renders.

— Keeper K701, 2026-07-15. FK exponent DERIVABLE: leptons are spinors $\rightarrow$ weight = scalar Szegő ($5/2$) – spin ($1/2$) = $E_0 = 2$ (=Elie's A, =the E-ladder, target-innocent). Needs Lyra's spinor-shift check, NOT the FK citation (which confirms). α CONDITIONAL PASS: 4π = the descent's 3D Coulomb solid angle (no free knob; identified, not Coulomb-re-derived). 6 params behind the $E_0=2$ check. See [[Keeper_K700...]], [[Keeper_K699...]], [[Keeper_K697...]].